MAT215: Complex Variables & Laplace Transformations

Quiz-04

January 11, 2026

Total - 20 Marks

(You need to answer **All questions**)

Name:

ID:

Section:

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1. Evaluate the following integral using cauchy integral theorem:

$$\int_{|z|=3} \frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-1)(z-2)} dz$$

Answer: Use the Residue Theorem to solve this math. (10 Marks)

2. Compute

$$\int_C \frac{1}{z} dz$$

Where C is a square centred at the origin with the vertices $\pm 1 \pm i$.
Answer:
First Method:
Here, the best option is the Cauchy integral. Let's rewrite the expression:

$$\int_C \frac{1}{z-0} dz = 2\pi i \underbrace{f(0)}_{1} = 2\pi i$$

Second Method:
Now, the worst method will be use the parametrization technique to solve this math. Sorry, but the risk level is too high, and I'm still unmarried.

3. Compute

$$\int_{|z|=3} \frac{z}{z^2+4} dz$$

Answer:
First Method:
Use the Residue theorem by rewriting the expression:

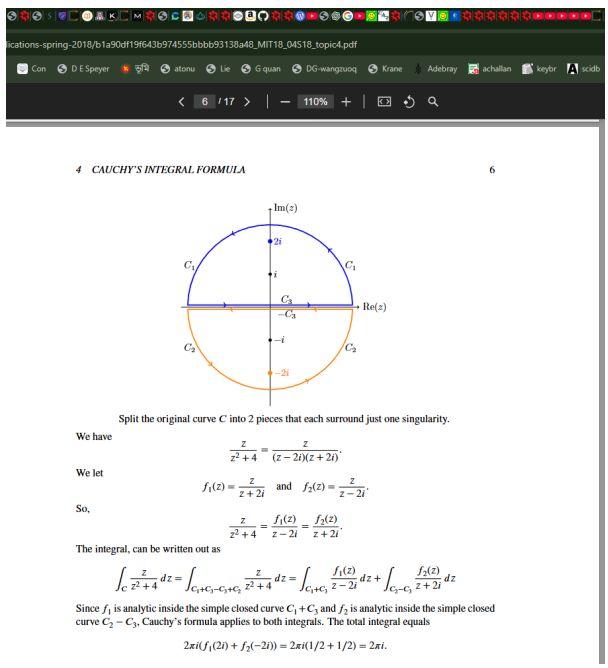
$$\frac{z}{z^2+4} = \frac{z}{(z+2i)(z-2i)}$$

$$\text{Res}(f, 2i) = \lim_{z\to 2i}\left[(z-2i)\frac{z}{(z-2i)(z+2i)}\right]$$

$$\text{Res}(f, -2i) = \lim_{z\to 2i}\left[(z+2i)\frac{z}{(z-2i)(z+2i)}\right]$$

$$\int_C \frac{z}{z^2+4}dz = 2\pi i(\text{Res}(f, 2i) + \text{Res}(f, -2i))$$

Second Method:
You can use the Cauchy theorem to solve this math as I illustrated in class: You



can have the full solution on page-6 from here: `https://ocw.mit.edu/courses/18-04-complex-variables-with-applications-spring-2018/b1a90df19f643b974555bbbb93138a MIT18_04S18_topic4.pdf`. (10 Marks)

Bonus Question:

1. Compute the integral

$$\int_{|z|=1} \frac{e^{z^2}}{z-2}dz$$

(1 Marks)

1 Formulas

Cauchy's Integral Formula:

Let $f(z)$ be analytic inside and on a simple closed curve C and let $z = z_0$ be any point inside C. Then,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint \frac{f(z)}{(z - z_0)^{n+1}} dz$$

Residue Theorem:

Let $f(z)$ be single-valued and analytic inside and on a simple closed curve C except at the singularities $z_1, z_2, z_3, \ldots, z_k$ inside C. Then the residue theorem states that:

$$\oint f(z)dz = 2\pi i \sum_{i=1}^{k} \mathrm{Re}\,(z = z_i)$$

where the Residue $\mathrm{Re}\,(z = z_i)$ can be calculated by,

$$\mathrm{Re}\,(z = z_i) = \lim_{z \to z_i} \frac{1}{(m-1)!} \frac{d^{(m-1)}}{dz^{(m-1)}} \left\{ (z - z_i)^m f(z) \right\}$$

where $z = z_i$ is the Pole of order m.

Best of Luck!